Bengaluru Traffic Pulse – Traffic Congestion Analysis

1. Project Overview

Bengaluru Traffic Pulse is a data analytics project that studies traffic congestion patterns across selected areas of Bengaluru. The project uses traffic-related data to identify areas with higher congestion, compare traffic conditions, and group areas using K-Means clustering.

The project also proposes two practical interventions to help improve traffic flow.

2. Problem Statement

Bengaluru experiences traffic congestion in several areas. Understanding differences in traffic volume, average speed, congestion level, road capacity utilisation and incident reports can help identify areas that may need traffic-management attention.

This project analyses the available dataset to identify congestion patterns and suggest possible interventions.

3. Project Objectives

- Clean and validate the Bengaluru traffic dataset.
- Analyse traffic conditions across different areas.
- Compare congestion, traffic volume, speed and incident reports.
- Group areas using K-Means clustering.
- Build an interactive Power BI dashboard.
- Recommend two traffic-management interventions.

4. Dataset

Dataset: Bengaluru Traffic Dataset
Source: https://www.kaggle.com/datasets/preethamgouda/banglore-city-traffic-dataset

- Records: 8,936
- Columns: 16
- Areas covered: 8
- Date range: 1 January 2022 to 9 August 2024

The dataset includes traffic volume, average speed, travel time index, congestion level, road capacity utilisation, incident reports, weather conditions and other traffic-related variables.

Please check the dataset's licence and terms before redistributing the original data.

5. Tools and Technologies

- Python
- Google Colab
- Pandas and NumPy
- Matplotlib
- Scikit-learn
- SQL / data analysis concepts
- Power BI
- GitHub

6. Data Cleaning and Quality Checks

The dataset was checked for:

- Missing values
- Duplicate records
- Invalid dates
- Negative numeric values
- Extra spaces in text fields

The original dataset contained 8,936 rows and 16 columns. No missing values or duplicate full rows were found. Date validation identified no invalid dates, and no negative values were found in the numeric columns.

A cleaned dataset was exported as `Bengaluru_Traffic_Cleaned.csv`.

7. Exploratory Data Analysis

Area-wise analysis identified the following average congestion levels:

Area	Average Congestion
Koramangala	93.99
M.G. Road	90.58
Indiranagar	87.64
Hebbal	80.09
Jayanagar	77.00
Whitefield	69.05
Yeshwanthpur	62.37
Electronic City	54.45

Koramangala recorded the highest average congestion, while Electronic City recorded the lowest among the eight areas in this dataset.

8. K-Means Clustering

K-Means clustering was performed using five area-level features:

- Average traffic volume
- Average speed
- Average congestion
- Average road capacity utilisation
- Average incident reports

The features were standardised using `StandardScaler`. Values of K from 2 to 5 were tested using silhouette scores.

K=2 produced the highest silhouette score among the tested values, approximately 0.551.

The resulting groups were interpreted as:

Higher-congestion group: Koramangala, M.G. Road, Indiranagar, Hebbal and Jayanagar.

Lower-congestion group: Whitefield, Yeshwanthpur and Electronic City.

These clusters are exploratory because the analysis uses only eight area-level observations.

9. Power BI Dashboard

The Power BI dashboard includes:

- Average congestion KPI
- Average speed KPI
- Average traffic volume KPI
- Total number of areas
- Area-wise congestion comparison
- Cluster-wise congestion comparison
- Weather-wise congestion analysis
- Area-wise incident reports
- Date and roadwork filters

10. Recommended Interventions

Intervention 1: Traffic Signal Optimisation

Prioritise high-congestion locations in Koramangala, M.G. Road and Indiranagar. Review signal timings at busy intersections and adjust them based on observed traffic demand.

Expected impact: Potential reduction in waiting time and queue build-up. The actual impact would need to be measured through a before-and-after evaluation.

Intervention 2: Peak-Hour Traffic Management

Prioritise bottlenecks in Koramangala, M.G. Road, Hebbal and Jayanagar. Consider managing turning movements and deploying traffic personnel during peak periods.

Expected impact: Potential improvement in traffic flow and reduction in congestion at selected bottlenecks. The actual impact would need field evaluation.

11. Limitations

- The dataset covers only eight named areas.
- Clustering uses eight area-level observations, so the results are exploratory.
- Congestion and capacity utilisation values include many observations at 100, which may indicate capped values.
- The analysis identifies patterns and associations; it does not establish causes of congestion.
- The interventions are recommendations, not measured or guaranteed improvements.
- Further field data and before-and-after evaluation are required to estimate actual intervention impact.

12. AI Use Disclosure

AI assistance was used to support code explanation, debugging, documentation structure and presentation preparation. The dataset analysis, generated outputs and project conclusions should be checked by the project team before submission.

13. Conclusion

The analysis identified clear differences in traffic congestion across the eight areas. Koramangala, M.G. Road and Indiranagar showed the highest average congestion, while Electronic City showed the lowest.

K-Means clustering grouped the areas into higher- and lower-congestion patterns. The findings support prioritising signal optimisation and peak-hour traffic management for further investigation.

The project demonstrates how data analytics, clustering and dashboard visualisation can support traffic-management decisions.